PSYKED.APP

Brain Performance Blueprint

45 Protocols on training 9 Brain Regions
to Unlimited Performance.

Science-backed performance neuroscience

by **The Psyked Team**

psyked.app

CONTENTS

Table of Contents

↓ Introduction: The Most Powerful Tool You Own 4

PART 1 — THE COMMAND CENTER

01 Prefrontal Cortex
The Willpower Engine 6

02 Anterior Cingulate Cortex
The Grit Circuit 10

03 Basal Ganglia
The Habit Automation System 13

PART 2 — THE LEARNING MACHINE

04 Hippocampus
The Learning Accelerator 17

05 Temporal Lobe
The Coaching Receiver 20

PART 3 — THE BODY-BRAIN INTERFACE

06 Cerebellum
The Precision Engine 23

07 **Amygdala**
The Threat Processor 27

08 **Brain Stem**
The Foundation 30

PART 4 — THE GATEKEEPER

09 **Thalamus**
The Attention Architect 34

PART 5 — STACKING THE PROTOCOLS

10 **The Complete System**
Daily, Pre-Competition & Recovery Stacks 38

BACK MATTER

★ **Quick Reference Cards**
All 9 regions · 45 protocols 42

★ **Recommended Reading**
10 essential books to go deeper 46

INTRODUCTION

The Most Powerful Performance Tool You Own — And Never Learned to Use

> Most athletes will never reach their ceiling. Not because of their body — because of their brain.

You've optimized your training. You've dialed your nutrition. You track HRV, VO2 max, sleep stages. You hire coaches, buy gear, study periodization. But the one organ that decides whether you push through or quit, whether you focus or drift, whether you choke under pressure or execute flawlessly — your brain — gets almost zero direct attention. That's the gap this book closes.

The sports world has spent decades perfecting the body. Nutrition science tells you what to eat. Training science tells you how to move. Recovery science tells you when to rest. But performance neuroscience — the science of how your brain actually drives everything — has been locked in academic journals. It hasn't reached the locker room, the gym floor, or the biohacker's morning routine. Not in a usable form. Until now.

> **Here's what most performance advice misses: everything is brain-first.** Discipline isn't a character trait — it's a function of your prefrontal cortex. Grit isn't a personality type — it's your anterior cingulate cortex under pressure. Your habits aren't choices — they're automated loops running through your basal ganglia.

Once you understand which brain regions drive which performance qualities — and how to train them — the whole game changes. That's what this book delivers.

How to use this book: Each chapter covers one brain region. You'll learn what it is, what it does, why it matters for performance, and the five evidence-based protocols you can implement immediately. Read front to back for the full picture, or jump to the region most relevant to your current bottleneck. Chapter 10 ties it all together into daily, pre-competition, and recovery stacks.

THE 9 REGIONS AT A GLANCE

01 **Prefrontal Cortex**
Willpower, decisions, discipline

02 **Anterior Cingulate Cortex**
Grit, effort, pain tolerance

03 **Basal Ganglia**
Habit automation, dopamine loops

04 **Hippocampus**
Memory formation, neurogenesis

05 **Temporal Lobe**
Coaching uptake, skill storage

06 **Cerebellum**
Motor precision, muscle memory

07 **Amygdala**
Stress response, pressure performance

08 **Brain Stem**
Autonomic baseline, HRV, breathing

09 **Thalamus**
Attention gating, sensory focus

Let's build your blueprint.

PART ONE

The Command Center

Prefrontal Cortex · Anterior Cingulate Cortex · Basal Ganglia

CHAPTER 01

Prefrontal Cortex

The Willpower Engine

> If your discipline collapses mid-session, your focus fractures under pressure, or your decision-making deteriorates late in competition — your prefrontal cortex is where you fix it first.

What It Is

The prefrontal cortex (PFC) is the most distinctly human part of your brain. Sitting right behind your forehead, it took evolution 200 million years to build. It's your brain's CEO — handling executive functions: planning, decision-making, impulse control, working memory, and the regulation of behavior in line with internal goals rather than immediate rewards. It's the part of you that says "I don't want to do this, but I'm going to do it anyway" — the entire foundation of high performance.

Why It Matters for Performance

The PFC is a metabolically expensive, chronically fatigued system. Every decision you make, every impulse you resist, every moment of sustained focus draws from the same limited resource pool. Roy Baumeister's ego depletion research showed that exercising self-control in one task measurably impairs self-control in a subsequent task. The fuel is finite.

Practically: a judge at 9am grants parole 65% of the time. The same judge at 4pm, after a full day of decisions? Under 10%. The cases didn't change. The brain did. Arrive to training with a depleted PFC and your discipline, focus, and pain tolerance are already compromised.

The Science

Research from Sara Lazar at Harvard showed regular meditators have measurably thicker prefrontal cortices than non-meditators — more gray matter means more processing capacity, better regulation, better performance. The PFC remains plastic — structurally changeable — throughout life. This is not fixed hardware. It is trainable tissue.

5 Protocols

1 Cold Exposure — 2–3 Minutes

> **Mechanism:** Cold water activates the locus coeruleus, releasing norepinephrine throughout the brain — sharpening PFC signal clarity. A 2022 study found 11 min/week of cold immersion significantly elevated norepinephrine. Equally: stepping into cold water is a PFC-over-amygdala rep. You're training the willpower circuit directly.

- → End your shower with 2–3 min cold (as cold as your shower gets)
- → Start with 30 sec and build over 2 weeks
- → Breathe slowly — resist the gasp reflex; this IS the training
- → Optimal timing: morning, to anchor alertness for the day
- → Advanced: ice bath at 57°F/14°C, 2–4 min, 3–4x per week

2 Daily Meditation — 10 Minutes

> **Mechanism:** Every time your mind wanders and you bring it back, you perform a bicep curl for your PFC. A 2011 Harvard study found measurable PFC gray matter increases after just 8 weeks of practice.

1. Sit comfortably, spine upright. Set a timer for 10 minutes.
2. Close your eyes, breathe naturally through your nose.
3. Focus attention on the sensation of breath at your nostrils.
4. When your mind wanders — notice it without judgment, return to breath.
5. That noticing-and-returning IS the practice. Every rep builds the PFC.

3 Decision Batching

> **Mechanism:** Every decision depletes the same PFC resource pool. Batch decisions to preserve capacity for high-stakes moments. This is why Jobs wore the same turtleneck — not rigidity, but neuroscience.

- → Sunday evening: decide training schedule, meals, and priorities for the week
- → Each morning: write your 3 most important tasks — don't renegotiate mid-day
- → Template small decisions: same breakfast, gear laid out the night before
- → Guard the first 2 hours: highest-value cognitive work before fatigue accumulates
- → No phone for the first 30 minutes of the day

4 Sleep Optimization — 7–9 Hours

> **Mechanism:** The PFC is the most sleep-sensitive brain region. A single night under 6 hours impairs PFC function comparable to legal intoxication. During deep sleep, the glymphatic system clears neurotoxic waste — this is maintenance, not just recovery.

- → Same wake time ±30 min, 7 days/week — anchors your circadian rhythm
- → Bedroom at 65–68°F (18–20°C) — core temp must drop for deep sleep
- → Block blue light 90 min before bed; bright sunlight within 30 min of waking
- → No caffeine after 12–1pm (half-life 5–7 hours)
- → 20–30 min deliberate wind-down: reading, stretching, journaling

5 Progressive Discipline Training

Mechanism: Just as muscle responds to progressive overload, the PFC responds to progressive challenge. Each impulse override strengthens synaptic connections between PFC and limbic system. Long-term consistent practice increases overall self-regulatory capacity.

- → Identify one small discomfort you avoid — do it daily for 30 days
- → Examples: cold shower, skipping dessert 5x/week, hardest task first each morning
- → The specific behavior matters less than the consistent PFC-over-impulse practice
- → Gradually escalate difficulty after each 30-day block
- → Track your streak — implementation intentions improve follow-through

CHAPTER SUMMARY

Prefrontal Cortex — 5 Protocols at a Glance

#	PROTOCOL	MECHANISM	ACTION
1	Cold Exposure	Norepinephrine + willpower reps	2–3 min cold shower daily
2	Meditation	Gray matter growth, attentional reps	10 min focused attention daily
3	Decision Batching	Preserve PFC for high-stakes moments	Weekly + morning planning ritual
4	Sleep	Glymphatic clearance, PFC recovery	7–9h, consistent schedule
5	Progressive Discipline	Synaptic strengthening	Daily small discomfort challenge

"Willpower is a biological resource — depletable, restorable, and trainable. Treat it like one."

KEY TAKEAWAY: Cold exposure, daily meditation, and consistent sleep are your three highest-leverage PFC protocols. Start with two minutes of cold water at the end of your next shower.

CHAPTER 02

Anterior Cingulate Cortex

The Grit Circuit

> If you quit too early in training, avoid the hardest reps, or fall apart when competition gets painful — your ACC is the ceiling you need to raise.

What It Is

Deep in the fold where your two brain hemispheres meet sits a thumb-sized strip of cortex that may be the most important region for competitive performance that nobody talks about: the anterior cingulate cortex (ACC). It's your brain's conflict monitor — evaluating the gap between what you want to do and what you need to do. Its most critical function: **it processes effort and pain.** Every time training becomes genuinely hard, the ACC runs the calculation: how much of this discomfort actually needs to stop you? This is the neurological home of grit.

The Pain/Effort Intersection

Every pain signal has two components: the **sensory component** (actual intensity) and the **affective component** (how threatening you perceive it). Elite athletes don't feel less physical discomfort — their ACC has become more effective at reducing the affective weight of that discomfort. They hurt just as much. But the hurt controls their behavior less.

The 40% Rule: Neuroscience

David Goggins' "40% rule" has a neurological basis. The ACC calculates a constant cost-benefit ratio for effort. A 2016 University of Birmingham study found that 6 weeks of maximal effort intervals changed not just physiology but psychological effort tolerance —

participants rated the same intensity as less unpleasant after the training block. The ACC is plastic. Train it.

1 High-Intensity Training

Mechanism: The ACC requires genuine discomfort to adapt. Zone 5 max-effort intervals repeatedly expose the ACC to discomfort and train its quit threshold upward — not just physiologically, but psychologically.

- 2x per week: 4–6 rounds of 30-sec all-out effort (sprint, assault bike, row)
- 2–3 min recovery between rounds
- During intervals: stay *with* the discomfort — don't mentally escape. This is the ACC training.
- Add 1 interval every 2 weeks as tolerance rises

2 Mindfulness for ACC Thickening

Mechanism: Long-term meditators have measurably thicker ACC cortex — and report lower pain unpleasantness at the same intensity. Meditation physically thickens the part of your brain that controls how much discomfort governs your behavior.

- 10–20 min open-monitoring meditation daily
- Allow thoughts and sensations to arise and pass without reacting
- When discomfort arises — observe it without acting on it
- This directly mirrors competition: observe discomfort without obeying it

3 Goal-Setting Rituals

Mechanism: The ACC calculates effort against goal-value. Higher perceived goal importance = higher effort ceiling. A 2017 meta-analysis found specific, challenging goals improved performance 12–15% vs. vague ones.

- → Daily: write your #1 training goal before each session — one specific sentence
- → Weekly: reconnect to your larger 'why' in writing
- → Pre-competition: 2–3 min ritual that reconnects you to purpose
- → Consistent rituals reduce anxiety and prime ACC for higher effort output

4 Caffeine Optimization — 200–400mg

Mechanism: Caffeine blocks adenosine receptors in the ACC, delaying perception of effort and fatigue. A meta-analysis of 34 studies found caffeine improves endurance by ~11% — primarily through reducing perceived exertion at the same workload.

- → Dose: 3–6mg/kg bodyweight (200–400mg for most athletes)
- → Timing: 30–60 min before performance for peak blood levels
- → Cycle caffeine: 5 days on, 2 off — or 5–7 day breaks every 4–6 weeks to restore sensitivity
- → Caution: above 400mg increases anxiety in many — the amygdala activates too

5 Iron Level Monitoring

Mechanism: Iron deficiency — even without anemia — reduces dopamine synthesis, lowering ACC motivational drive and raising perceived exertion. A 2004 study found iron-deficient non-anemic women had significantly higher RPE at the same workload — normalized after supplementation.

- → Get ferritin tested (not just hemoglobin) 2x per year
- → Optimal athlete ferritin: 50–100+ ng/mL (labs flag >12 as 'normal' — not optimal)
- → Iron-rich foods: red meat, shellfish, organ meats, legumes + vitamin C
- → Test before supplementing — over-supplementation is harmful

CHAPTER SUMMARY

The ACC is where your ceiling lives.

#	PROTOCOL	ACTION
1	HIIT	2x/week max-effort intervals; stay with the discomfort
2	Mindfulness	10–20 min open monitoring daily
3	Goal Rituals	Daily purpose statement before every training session
4	Caffeine	200–400mg, 30–60 min pre-performance; cycle regularly
5	Iron Monitoring	Ferritin test 2x/year; target 50–100+ ng/mL

"The 40% rule isn't motivation. It's your ACC. Train it."

KEY TAKEAWAY: Your effort ceiling is not fixed — it's a trainable threshold. Add one 30-second all-out interval to your next session and stay with the discomfort instead of backing off.

CHAPTER 03

Basal Ganglia

The Habit Automation System

> If training consistency feels like constant willpower battles, or stress derails routines you've built for months — your basal ganglia automation system is under-trained.

What It Is

Deep beneath your cerebral cortex, the basal ganglia are your brain's automation engine. They don't think or plan — they run programs. Deeply learned behavioral sequences execute automatically based on environmental cues and expected rewards. Your habits live here, running on dopamine.

The Dopamine Loop

The habit loop: **cue → routine → reward.** With repetition, your basal ganglia absorb behaviors from your PFC and automate them. Wolfram Schultz's landmark studies showed dopamine fires not at reward delivery but in *anticipation* of it — shifting toward the cue. A habit requires only a cue. A decision burns PFC energy. Habits win when willpower is depleted — every time.

Why Habits Are Harder to Break Than You Think

Bad habits don't get erased — they get competed with. Under stress, PFC control diminishes and basal ganglia automaticity takes over. The deepest groove wins by default. Don't try to break bad habits through willpower alone. Build competing habits so deeply they win even when the PFC is offline.

1 Habit Stacking

Mechanism: Anchor new habits to existing ones. The existing habit provides the cue — the basal ganglia adopt the new behavior faster because the triggering infrastructure already exists.

- → "After I pour morning coffee, I will write my 3 priorities"
- → "After I sit to change shoes at the gym, I will do 2 min mobility"
- → "After I finish training, I will immediately drink my recovery shake"
- → "After I brush teeth, I will take magnesium glycinate"
- → Specificity is everything. Vague intentions fail. Precise behavioral anchors succeed.

2 Reward Timing — Immediate Celebration

Mechanism: Dopamine must fire immediately after the behavior — not hours later. The basal ganglia care about the 3-second window. B.J. Fogg's research found immediate celebration dramatically accelerates habit formation.

- → Celebrate immediately after each target behavior rep: fist pump, verbal 'done,' felt satisfaction
- → For training: celebrate after completing hard sets, intervals, or full sessions
- → A tiny celebration gets the same dopamine reinforcement as a large one. That's the point.

3 Dopamine Hygiene

Mechanism: Large dopamine spikes from cheap sources (social media, pornography, sugar) recalibrate the reward system downward — making genuine achievement feel less rewarding. Protect your dopamine baseline before training.

- → No social media scrolling in the first 30 min after waking
- → Save highest-reward activities for after your most important work
- → Let anticipation of the reward drive effort — don't pre-load the reward before effort begins

4 Schedule Consistency

Mechanism: Same training time, same place, same ritual = powerful cue structure the basal ganglia use to pre-initiate training behavior automatically — with less PFC drag.

- → Fixed training time, protected like your most important appointment
- → Consistent pre-training ritual: same music, warm-up, mental prep sequence
- → Train in the same location when possible
- → When schedule changes, recreate as many ritual elements as possible

5 Diet for Dopamine — Tyrosine-Rich Foods

> **Mechanism:** Dopamine is synthesized from tyrosine. A 2015 meta-analysis found tyrosine improved cognitive flexibility under high-stress, high-demand conditions — precisely when dopamine gets depleted.

- → High-tyrosine foods: chicken, turkey, eggs, fish, beef, pumpkin seeds, almonds
- → Include tyrosine-rich protein source in your pre-training meal
- → L-tyrosine supplement: 500–2000mg, 30–60 min pre-performance on empty stomach
- → Pair with B vitamins (B6, B9, B12) — required cofactors for dopamine synthesis

CHAPTER SUMMARY

Your habits are not your character — they're your architecture.

"Stack habits. Celebrate immediately. Guard your dopamine. Stay consistent. Feed the system."

KEY TAKEAWAY: Your most powerful habit change isn't willpower — it's architecture. This week, attach one new performance behavior to an existing daily habit using the "After X, I will Y" format.

PART TWO

The Learning Machine

Hippocampus · Temporal Lobe

CHAPTER 04

Hippocampus

The Learning Accelerator

> If coaching doesn't stick, skills take longer to automate than they should, or you plateau despite consistent effort — your hippocampus is the learning accelerator to upgrade.

What It Is

Curled deep in the medial temporal lobe, the hippocampus is your brain's primary memory-formation hub. Your brain generates new neurons here throughout your lifetime — neurogenesis — and we know exactly what controls it. Your hippocampus converts short-term experiences into long-term memories during sleep, and new neurons improve pattern recognition, cognitive flexibility, and coaching uptake.

Why Aerobic Exercise Literally Grows Your Brain

In 1999, researchers discovered running mice had measurably more new hippocampal neurons than sedentary mice. A 2011 study by Erickson et al. confirmed this in humans: 6 months of aerobic exercise increased hippocampal volume by 2% — reversing 1–2 years of age-related shrinkage. The mechanism: aerobic exercise dramatically elevates Brain-Derived Neurotrophic Factor (BDNF) — "Miracle-Gro for the brain." Exercise is the most potent natural trigger for it.

1 Zone 2 Cardio for BDNF

Mechanism: Zone 2 aerobic exercise (60–70% max HR) produces the highest sustained BDNF output. John Ratey calls this the 'sweet spot' for neurogenesis. High-intensity work produces adrenaline that partially suppresses hippocampal BDNF — Zone 2 keeps the signal elevated throughout.

- → 3–4 sessions per week, 30–60 min at Zone 2 (60–70% max HR)
- → Talk test: full sentences, slightly effortful
- → Learn immediately after Zone 2 — BDNF stays elevated 30–60 min post-exercise
- → Best modalities: running, cycling, swimming, rowing

2 REM Sleep Optimization

Mechanism: REM sleep is when the hippocampus replays the day's experiences to the cortex, consolidating learning. Cut REM short, and the previous day's learning is incomplete. EEG studies confirm hippocampal replay sequences during REM mirror waking learning sequences.

- → Protect the last 2 hours of sleep — REM concentrates in final sleep cycles
- → No alcohol: even 2 drinks reduces REM by 20–25%
- → Cool, dark, quiet sleeping conditions for optimal sleep architecture
- → Sleep a full night immediately after learning critical material — that night matters most
- → 20-min post-learning nap also boosts hippocampal consolidation

3 Omega-3/DHA Supplementation

Mechanism: DHA is the most abundant omega-3 in the brain, essential for hippocampal membrane fluidity and neurogenesis. Low DHA reduces BDNF expression and impairs memory formation. A 2012 study found DHA supplementation increased BDNF and improved spatial learning.

- → Target: 1–3g combined EPA+DHA daily
- → Fatty fish (salmon, sardines, mackerel) 2–3x/week
- → Supplement: high-quality fish oil or algae-based omega-3
- → Take with fat-containing meals for better absorption
- → Allow 4–8 weeks for measurable cognitive effects

4 Novel Skill Learning

Mechanism: Your hippocampus is most active during genuinely new information acquisition. Repetition of learned skills primarily activates the basal ganglia. Novelty keeps hippocampal neurogenesis robust and cognitive flexibility sharp.

- → Commit to one genuinely new skill per quarter — outside your current expertise
- → For athletes: cross-train in a different sport or add an entirely new skill element
- → Key variable: genuine challenge and novelty — comfortable familiar activity doesn't drive the same hippocampal engagement
- → **HOW TO START TODAY:** Schedule a single 60-min session this week in a skill you've never tried. Book it now — decision latency kills follow-through.

5 Alcohol Awareness

Mechanism: Alcohol suppresses BDNF, disrupts REM sleep, and with chronic use directly damages hippocampal neurons. A 2017 BMJ study found even moderate drinking (14–21 units/week) was associated with hippocampal atrophy.

- → No alcohol the night before important learning or skill training sessions
- → In learning-intensive training blocks: 0–2 drinks/week maximum
- → Even 2 drinks reduces consolidation of content learned that day
- → This is pure performance — alcohol and learning are antagonists

CHAPTER SUMMARY

Your hippocampus is both a memory bank and a living factory.

"Exercise grows your brain. Sleep seals the learning. Feed it DHA. Challenge it with novelty. Protect it from alcohol."

KEY TAKEAWAY: Aerobic exercise is the most powerful tool you have for growing your learning capacity. Do 30 minutes of Zone 2 cardio this week, then immediately review a skill or coaching point while your BDNF is elevated.

CHAPTER 05

Temporal Lobe

The Coaching Receiver

> If coaching takes multiple sessions to stick, your self-talk undermines execution, or you struggle to retain verbal instructions under pressure — your temporal lobe processing speed is the bottleneck.

What It Is

Stretching along the sides of your brain, the temporal lobes handle language, auditory processing, semantic memory, and complex sensory integration. For athletes, the temporal lobe is your **coaching receiver** — determining how efficiently you extract, process, and retain information from coaches, instructors, and your own self-talk. The path from verbal instruction to motor learning runs directly through this region.

How Elite Athletes Absorb Coaching Faster

A 2014 study comparing elite and recreational tennis players found that experts showed faster neural responses to both auditory and visual cues — their temporal lobe processing was faster and more selective. Elite performers don't just have better physical conditioning. They have better coaching uptake. This temporal lobe efficiency is trainable.

1 Active Listening Techniques

Mechanism: Active listening activates the full temporal network including Wernicke's area, hippocampus, and PFC. Deeper processing creates stronger neural traces and dramatically better retention than passive hearing.

- → During coaching: phone away, 100% attentional focus
- → After each instruction: paraphrase in your own words before acting
- → Ask one clarifying question per major coaching point
- → Post-session: within 5 min, write the 2–3 key coaching points

2 Music and Rhythm Training

Mechanism: Musical training produces measurable temporal lobe structural changes. Beat synchronization: music at appropriate BPM improves endurance performance by 10–15% through motivational and temporal entrainment effects.

- → Train with rhythmically appropriate music matching effort level (120–170 BPM)
- → Sport-specific: rhythm drills, metronome work, cadence training
- → Even 15 min/day of any instrument builds temporal lobe networks
- → During skill drills: count rhythms or use verbal timing cues

3 Verbal Rehearsal and Self-Talk

Mechanism: Inner speech is processed through the same temporal networks as external speech. A meta-analysis of 47 studies found instructional self-talk reduced technical errors by 15–25% in technique-dependent tasks.

- → Develop 1–2 personal cue words for each key technical skill
- → Use instructional cues during practice reps ('loose,' 'drive,' 'smooth')
- → Pre-performance verbal sequence: one specific execution-focused statement
- → Under pressure: cue words re-engage the temporal-to-motor pathway

4 Audiobooks and Podcasts for Passive Priming

Mechanism: Expert audio builds conceptual schemas in temporal semantic memory. Richer schemas mean new coaching information slots in faster — accelerating learning uptake.

- → 1–2 hrs/week: coaching breakdowns, performance science, sport-specific experts
- → Commute listening is ideal timing
- → After listening: write one sentence summarizing the key insight
- → Huberman Lab, sport-specific coaching podcasts, expert interview series

5 Social Interaction as Workout

Mechanism: Face-to-face interaction simultaneously processes voice, expression, and emotional tone — one of the most complex temporal lobe tasks. Regular high-quality social interaction maintains temporal processing speed and emotional intelligence.

- → Prioritize in-person over digital communication when possible
- → Schedule 2–3 genuine social engagements per week
- → Practice high-quality conversation: real questions, active listening, no phone
- → Team debriefs and feedback sessions build temporal capacity AND accelerate learning
- → **HOW TO START TODAY:** Put your phone away for one full conversation today and actively paraphrase what the other person says. Temporal workout complete.

CHAPTER SUMMARY

The temporal lobe: decoder ring between information and performance.

"Listen actively. Train with rhythm. Talk yourself through it. Absorb expert knowledge. Stay socially sharp."

KEY TAKEAWAY: After your next coaching session, spend 5 minutes writing the 2-3 key points in your own words. That single habit doubles retention — starting today.

PART THREE

The Body-Brain Interface

Cerebellum · Amygdala · Brain Stem

CHAPTER 06

Cerebellum

The Precision Engine

> If your technique breaks down under fatigue, skills plateau despite high training volume, or movement feels mechanical under pressure — your cerebellum needs deliberate attention.

What It Is

At the back of your skull, the cerebellum houses more neurons than the rest of the brain combined. Its job: precision. It fine-tunes every movement by comparing what your motor cortex intended with what your proprioceptors report — correcting in real time with millisecond accuracy. It's the primary site of motor learning — the neural substrate of "muscle memory."

Why Deliberate Practice Beats Raw Hours

The cerebellum requires error signals to update its internal model. Repetition without feedback grooves incorrect patterns. This is why 10,000 hours of *deliberate practice* — near-edge-of-ability, with feedback — produces expertise, while 10,000 hours of mindless repetition does not.

Visualization Science

Vivid visualization activates the motor cortex and cerebellum in patterns that closely mirror actual execution. Pascual-Leone et al. (1995) found mental rehearsal of a piano sequence produced similar cerebellar changes as physical practice. Visualization is neurological training — not motivational psychology.

1 Deliberate Practice with Feedback Loops

Mechanism: Error signals drive cerebellar motor refinement. Every caught-and-corrected error is a cerebellar training stimulus.

- → Break skills into smallest components; work at edge of ability
- → Immediate feedback each rep: coach, video review, or measurable outcome
- → Target 70–80% execution accuracy — too easy = no challenge
- → 20–30 min of full-focus beats 2 hrs of distracted repetition
- → Video yourself — watching your own movement activates cerebellar error-correction

2 Proprioception Training

Mechanism: Cerebellar precision depends on the quality of proprioceptive input. Better sensory input = better cerebellar calibration = more precise movement output.

- → Single-leg balance: 3 sets x 30–60 sec per leg, progressing to eyes closed
- → Unstable surface training: BOSU ball, wobble board, barefoot on varied terrain
- → Agility ladder drills for rapid, precise foot placement challenges
- → 5–10 min balance work as warm-up, 3–4x/week

3 Mental Rehearsal / Visualization

> **Mechanism:** Vivid first-person visualization activates cerebellar motor programs during imagined execution. The more kinesthetically specific, the stronger the cerebellar training effect.

1. Find quiet space. Sit comfortably, close your eyes.
2. Relax with 3–5 slow breaths.
3. Visualize from first-person — feel the movement in your body, not watch from outside
4. Include all senses: sight, feel in muscles and joints, the sound of the movement
5. Visualize *successful* execution — the cerebellum builds the pattern you rehearse

4 Cross-Training for Motor Richness

> **Mechanism:** Athletes trained across multiple movement modalities have more sophisticated cerebellar maps — better balance, faster skill acquisition, superior injury resistance. The neurological basis of the 'athletic base.'

- → Include one different movement modality weekly: yoga, martial arts, gymnastics, dance, climbing
- → For specialized athletes: add rotation and overhead components if your sport lacks them
- → Consider annual 3–4 week cross-training blocks for cerebellar novelty
- → **HOW TO START TODAY:** Add one 30-min session this week in a modality completely different from your primary sport. Any novel movement counts — your cerebellum rewards variety immediately.

5 Rest Windows for Consolidation

Mechanism: Motor memory consolidation — transferring newly learned patterns to stable cerebellar engrams — happens during rest and sleep. Walker and Stickgold showed sleep-dependent consolidation accounts for 20–30% performance improvement with no additional practice.

- → Rest or low intensity the day after high-volume technical practice
- → Sleep quality determines motor consolidation — slow-wave sleep is critical
- → Don't stack multiple highly technical sessions without sleep between them
- → 'Offline' learning is real — sometimes you wake up better. That's the cerebellum.

CHAPTER SUMMARY

Quality reps, clear feedback, sensory richness, and rest to consolidate.

"Practice deliberately with feedback. Train balance. Visualize vividly. Cross-train. Sleep to consolidate."

KEY TAKEAWAY: Twenty focused minutes with immediate feedback beats two hours of mindless repetition. Tonight before bed, spend 5 minutes visualizing tomorrow's key skill in first-person — your cerebellum will consolidate it while you sleep.

CHAPTER 07

Amygdala

The Threat Processor

> If nerves kill your performance, you choke on the big stage, or overtraining leaves you anxious and burned out — your amygdala is running the show when it shouldn't be.

What It Is

Two almond-shaped clusters of neurons buried in the medial temporal lobe — the amygdalae are your threat detection system. When they detect danger (real or perceived), they trigger the stress response: cortisol, adrenaline, heart rate acceleration, and a cognitive shift from PFC-led rational thinking to rapid, survival-oriented behavior. In ancestral environments, lifesaving. In competitive performance environments, potentially catastrophic.

The Choking Problem

Choking under pressure is largely an amygdala phenomenon. High amygdala activation causes two problems: **PFC hijacking** (reduced working memory and decision quality) and **cerebellar bypass** (motor control shifts from smooth automatic cerebellar programs to slow, mechanical cortical control). The choke is literal — you're over-thinking movements that normally run automatically.

Train Your Stress Response — Don’t Eliminate It

Research by Wendy Mendes distinguishes **threat state** (demand exceeds perceived resources — performance degrades) from **challenge state** (demand is high but resources feel adequate — performance enhances). The goal isn't to eliminate stress — it's to shift

your default interpretation from threat to challenge, through building a learned history of handling difficult situations successfully.

1 Box Breathing — 4-4-4-4

> **Mechanism:** Slow, controlled breathing activates the vagus nerve, stimulates parasympathetic activity, reduces cortisol, and directly dampens amygdala firing. Used by Navy SEALs, surgeons, and elite athletes for high-pressure performance.

1. Exhale completely to empty your lungs
2. Inhale through your nose for 4 counts
3. Hold at the top for 4 counts
4. Exhale for 4 counts
5. Hold empty for 4 counts — repeat 4–6 cycles (about 2 minutes)

2 Progressive Stress Exposure

Mechanism: Every time you enter a high-stress situation and execute successfully, you build a cortical memory of handling it — reducing amygdala reactivity to similar situations in future. The amygdala learns from experience.

- → Create progressively challenging practice environments simulating competitive pressure
- → Time pressure, audience observation, fewer attempts, consequences for failure
- → Tournament simulation: practice under full competition conditions regularly
- → Gradually escalate stakes: intrasquad → local → regional → national

3 Chronic Stress Management

Mechanism: Chronic stress increases amygdala reactivity while simultaneously impairing PFC regulation — creating the vicious cycle of overtraining and burnout. The amygdala doesn't distinguish between training stress and life stress. Both drain the same bucket.

- → Track HRV daily — declining trend over 5–7 days indicates accumulated stress
- → When HRV drops 10%+ below 7-day average: prioritize recovery over training intensity
- → Active recovery: Zone 1 movement, massage, sauna
- → Social connection: oxytocin from positive interaction directly inhibits cortisol release

4 Journaling for Threat Externalization

Mechanism: Naming a threat in words reduces amygdala activation. Lieberman et al. (2007) showed that naming emotions on a face significantly reduced amygdala response. 'Affect labeling' is a measurable amygdala regulation tool. Journaling applies this to performance anxiety.

- → Pre-competition (10 min): write what you're feeling; name the fears specifically; then write your preparation, capability, and plan
- → Post-competition: what went well; what to adjust; what you're proud of
- → General stress journaling 10 min, 3x/week — getting stress out of the monitoring loop reduces baseline amygdala activation

5 Cold/Heat Contrast Training

Mechanism: Repeated temperature extremes provide a controlled physiological stressor the amygdala learns to regulate. With repeated exposure, the default response shifts: 'uncomfortable but not dangerous, and it ends soon.' This trains remaining cognitively functional while physiologically stressed.

- → Cold immersion (57°F/14°C): 2–4 min, 3–4x/week, building up from shorter durations
- → Sauna: 15–20 min at 170–190°F, immediately followed by cold
- → During cold: keep breathing slow and deliberate; be present in the discomfort
- → Advanced: perform cognitive tasks (arithmetic, verbal recall) during cold — trains PFC function under acute stress

CHAPTER SUMMARY

The amygdala needs sophisticated management for a performance context.

"Breathe deliberately. Expose progressively. Manage chronic load. Journal the fear. Embrace the cold."

KEY TAKEAWAY: Your goal isn't to eliminate stress — it's to shift from threat to challenge. Before your next high-pressure session, run 4-6 cycles of 4-4-4-4 box breathing and write one sentence about your preparation. Those two minutes rewire your stress response.

CHAPTER 08

Brain Stem

The Foundation

> If your HRV trends down during training blocks, you never feel truly recovered, or fatigue arrives earlier than your fitness should allow — your autonomic foundation needs rebuilding from the ground up.

What It Is

Beneath the cerebral cortex, your brain stem connects to your spinal cord and controls the autonomic functions that keep you alive: breathing, heart rate, blood pressure, sleep-wake cycles. Its activity never stops. Every performance system in your brain depends on the brain stem's autonomic baseline being stable. Fix the foundation, and everything above it works better.

HRV as Your Performance Biomarker

Heart Rate Variability (HRV) — the variation between successive heartbeats — is the most accessible window into brain stem autonomic function. High HRV = healthy parasympathetic/sympathetic balance. Low HRV = accumulated stress, reduced resilience. Track it daily with a wearable (WHOOP, Garmin, Oura) or free app. When HRV drops 10%+ below your 7-day average, prioritize recovery over intensity.

1 Nasal Breathing

Mechanism: Nasal breathing filters and humidifies air, produces nitric oxide that dilates airways, and drives a slower, diaphragm-led pattern that activates the parasympathetic nervous system via the vagus nerve. James Nestor's research showed 10 days of nasal-only breathing dramatically improved HRV, sleep quality, and endurance markers.

- → Practice nasal-only breathing during Zone 1–2 exercise
- → If you can't breathe nasally at a given intensity — you're going too fast for aerobic training
- → Sleep with your mouth closed — use medical-grade mouth tape if you're a chronic mouth breather
- → Become conscious of when you're mouth breathing during the day and switch back

2 Neck Mobility Work

Mechanism: The vagus nerve exits the skull through the neck. Restricted neck mobility and chronic forward head posture impair vagal tone. The densely proprioceptive suboccipital muscles create a constant sympathetic activation signal when chronically tense.

- → Daily 5-min circuit: chin tucks, lateral bends (ear to shoulder), rotation, suboccipital release (tennis ball against wall)
- → For every hour of desk work: 5 min chin tucks and thoracic extension
- → Avoid phone use with head down — 'text neck' at 30° multiplies effective head weight from 12 to 40 lbs

3 Sleep Schedule Consistency — Circadian Anchoring

Mechanism: The brain stem's suprachiasmatic nucleus (SCN) orchestrates every physiological process. The most powerful SCN anchor is light exposure — bright light in the first hour of waking, darkness in the final hours before sleep.

- → Same wake time ±30 min, 7 days/week — even weekends (social jet lag disrupts SCN)
- → Bright light exposure within 30 min of waking — go outside or use a 10,000-lux light therapy box
- → Darkness 90 min before bed: blackout curtains, blue-light glasses, dimmed screens
- → Eat your largest meals during daylight hours — meal timing supports circadian rhythm

4 Magnesium Glycinate

Mechanism: Magnesium is a cofactor in 300+ enzymatic reactions including ATP production and GABA activity. Athletes frequently have suboptimal levels due to sweat loss and high metabolic demand. A 2017 meta-analysis found magnesium supplementation significantly improved sleep quality and efficiency.

- Dose: 300–400mg elemental magnesium as magnesium glycinate per day
- Timing: 60 min before bed — the calming effect supports sleep onset and deep sleep
- Avoid magnesium oxide (cheap, poorly absorbed, causes GI distress)
- Food sources: dark leafy greens, pumpkin seeds, almonds, dark chocolate, black beans

5 Diaphragmatic Breathing / Vagus Nerve Activation

Mechanism: Deep diaphragmatic breathing physically stimulates vagal afferents with each breath cycle, generating parasympathetic activation throughout the body. The exhale is more potent than the inhale — extending the exhale maximizes vagal activation.

1. Inhale through your nose for 4 counts
2. Exhale slowly for 6–8 counts (continuous flow, no holding)
3. Continue for 5 minutes
4. Use post-training (accelerates parasympathetic recovery), pre-sleep, and between competition efforts

CHAPTER SUMMARY

The brain stem is the silent foundation of all performance.

"Breathe through your nose. Free your neck. Anchor your sleep schedule. Supplement magnesium. Activate your vagus nerve."

KEY TAKEAWAY: Your autonomic baseline determines your recovery ceiling. Tonight: take 300mg of magnesium glycinate 60 minutes before bed. This week: switch to nasal-only breathing during Zone 1-2 training. Two habits, massive foundation impact.

PART FOUR

The Gatekeeper

Thalamus

CHAPTER 09

Thalamus

The Attention Architect

> If distractions derail focus mid-session, noisy environments destroy your performance, or you can't sustain deep concentration when it matters most — your thalamic attention filter is the system to sharpen.

What It Is

Deep in the center of your brain, the thalamus is your brain's central switching station. Nearly all sensory information passes through it before reaching the cortex. Every moment, you receive millions of sensory signals — the thalamus decides which ones reach conscious awareness and which get filtered out. This makes it your attention gatekeeper.

Why Your Environment Determines Your Focus

The thalamus's filtering is controlled by top-down signals from the PFC. A calm, rested, meditation-trained brain filters distractions far more effectively than a tired, stressed one. A noisy, cluttered, notification-filled environment forces the thalamus to process and suppress millions of competing signals — consuming resources that would otherwise support deep focus. Design your environment, and you design your focus.

Sleep Spindles and Next-Day Performance

Sleep spindles — brief bursts of rapid neural oscillations generated by the thalamus during Stage 2 NREM sleep — are the primary mechanism for gating sensory disruptions during sleep AND for transferring motor and procedural memories to long-term storage. More

spindles = better consolidation = better next-day performance. They're maximized by consistent sleep schedules, cool sleeping environments, and magnesium.

1 Sleep Quality Optimization

> **Mechanism:** Thalamic sleep spindles require adequate Stage 2 and Stage 3 NREM sleep, maximized by consistent schedule, cool temperature, no alcohol, and a proper wind-down routine.

- → 65–68°F (18–20°C) bedroom temperature maximizes NREM quality
- → No alcohol within 4 hours of sleep — reduces spindle density by 20–30%
- → Ear plugs or white noise in noisy environments — protect thalamic sensory gating during sleep
- → No screens 60–90 min before bed — keeps thalamus in high-alert mode when it needs to downshift

2 Sensory Training Drills

Mechanism: The thalamus's filtering ability is trainable. Regular practice discriminating fine sensory detail develops more efficient thalamic filtering circuitry — faster extraction of relevant signal from noise. Part of why expert athletes 'see' the game differently.

- → Visual tracking: soft gaze scanning, reaction ball, multiple-object tracking drills
- → Auditory: deliberately attend to subtle environmental cues during training (footsteps, opponent breath)
- → Proprioceptive: eyes-closed balance work and movement replication
- → Dual-task training: perceptual task simultaneously with a physical skill — trains thalamic multisensory integration

3 Caffeine Timing — Adenosine Blocking

Mechanism: The thalamus has high adenosine receptor density — caffeine restores thalamic gating efficiency by blocking adenosine. Delaying the first dose 60–90 min after waking prevents the afternoon energy crash that occurs when you caffeinate immediately on waking.

- → Delay first caffeine dose 60–90 min after waking — allow the morning cortisol awakening response to clear first
- → Last dose by 1pm to protect sleep architecture that night
- → High-performance days: 100–200mg at 60–90 min post-wake, optional 100–200mg at midday if needed

4 Mindfulness for Top-Down Filtering

Mechanism: Mindfulness training strengthens PFC-to-thalamus signaling, improving thalamic filtering efficiency. Experienced meditators show measurably better sensory gating at the thalamic level compared to matched controls.

- → Focused Attention meditation (Chapter 1) also trains thalamic filtering as a secondary benefit
- → For competition-specific training: practice in mild distraction — coffee shop, gym lobby, outdoor spaces
- → Attentional breathing: during any activity, hold focus on one sensory channel for 30–60 sec while all else fades to background

5 Environmental Design — Sensory Hygiene

> **Mechanism:** The thalamus processes every sensory input it receives. A high-sensory-load environment taxes the filtering system continuously, consuming neural resources that would otherwise support deep focus. Design your environment, and you design your thalamic bandwidth.

- → Work in a clean, visually uncluttered environment — a clear desk supports clear thinking
- → Phone on DND / airplane mode during focus blocks — notifications are thalamic hijacks by design
- → Single-tab browsing during cognitive work — each open tab competes for processing
- → Before high-performance sessions: 2–3 min in genuine quiet with eyes closed — resets thalamic baseline
- → Train in environments that simulate competition — thalamic habituation reduces sensory load during performance

CHAPTER SUMMARY

Control what gets through to your mind and your focus becomes a precision instrument.

"Optimize sleep architecture. Train your senses. Time caffeine. Meditate for top-down control. Design your sensory environment."

KEY TAKEAWAY: Your environment is your focus system. Right now: phone on DND, clear your workspace, and do 2 minutes of silent eyes-closed breathing before the next thing on your agenda. That's thalamic reset in real time.

PART FIVE

Stacking the Protocols

The Complete System

CHAPTER 10

The Complete System

How the 9 Regions Work Together

The Architecture

Your brain is one deeply interconnected system where every region communicates with every other. The **brain stem** is the foundation — managing autonomic state, sleep stages, and breathing efficiency. The **thalamus** is the gateway — determining what sensory information reaches the cortex. The **amygdala** is the alarm — when over-activated, it hijacks the PFC and bypasses the cerebellum. The **PFC and ACC** are the directors — planning, regulating, and pushing through discomfort. The **basal ganglia** run your automated programs. The **hippocampus and temporal lobe** are your learning architecture. The **cerebellum** executes with precision.

Neglect any one, and performance leaks somewhere in the chain. Optimize the whole system, and gains compound on each other.

The Non-Negotiables

> **Sleep (7–9 hours, consistent schedule)** — Affects every single region. PFC glymphatic maintenance. ACC motivational recovery. Basal ganglia habit consolidation. Hippocampal memory encoding. Cerebellar motor consolidation. Amygdala emotional reset. Brain stem HRV restoration. Thalamic spindle generation. A single night of inadequate sleep costs you everywhere. There is no protocol in this book that compensates for chronic sleep debt.

Aerobic Exercise (4–5 hours/week) — Increases BDNF for hippocampal neurogenesis. Elevates norepinephrine for PFC clarity. Regulates dopamine for basal ganglia motivation. Reduces baseline cortisol protecting the amygdala. Improves vagal tone for brain stem HRV. Enhances thalamic gating via sleep quality improvement. If you do nothing else in this book: sleep properly and exercise consistently. Everything else is a multiplier.

Morning Protocol Stack

6am **Wake — Consistent Time**

> The morning sets the neurological tone for the entire day.

- → Immediate: out of bed, no phone for first 30 minutes
- → 5 minutes outside or bright window — sunlight for SCN anchoring
- → Cold shower ending (2–3 min cold) — norepinephrine spike, dopamine tone
- → 10 min meditation — PFC + thalamic filtering training
- → Write 3 priorities + today's training goal (decision batch + ACC prime)
- → Tyrosine-rich breakfast: eggs, meat, seeds, fish
- → First caffeine at 60–90 min post-wake (200mg)

Pre-Competition Stack

-90min

Pre-Performance Protocol

> Prime every relevant system before the moment of performance.

- 90 min out: review preparation notes; reconnect to your 'why' (ACC priming)
- 60 min out: caffeine dose (3–6mg/kg); begin physical warm-up
- 30 min out: consistent pre-performance ritual (music, movement, mindset cues); mental rehearsal in first-person
- 5 min out: box breathing — 4–6 cycles of 4-4-4-4
- Final: one verbal cue word connecting to technical intent, not outcome fear
- Self-talk: 'I've prepared for this. I execute.'

Recovery Stack

Post

Post-Performance Recovery Protocol

> The performance period continues after you stop competing.

- → Immediately: 5 min diaphragmatic breathing (in 4, out 6–8) — parasympathetic activation
- → Celebrate your effort immediately — dopamine reinforcement of the training loop
- → Protein + carbohydrate within 30 minutes
- → That evening: 10 min journaling — what went well, what to adjust, what you're proud of
- → Magnesium glycinate 300–400mg (60 min before bed)
- → No screens 90 min before bed; full sleep cycle for complete consolidation

Common Mistakes

Overtraining

High physical volume + high cognitive demands simultaneously deplete PFC, ACC, and brain stem resources. Build recovery weeks into every training block. Watch HRV trends. Recovery is when adaptation happens — not during training.

Sleep Debt

There is no hack around it. Caffeine masks symptoms but doesn't restore function. Consistently sleeping under 7 hours: fix this first. Everything else is cosmetic until you do.

Cheap Dopamine

Social media, pornography, and sugar before training blunt the dopamine drive that makes training feel rewarding. Dopamine hygiene is non-optional for anyone serious about sustained motivation.

Skipping Fundamentals

Advanced protocols are multipliers. They multiply what's already there. If sleep, exercise, and consistent practice aren't in place — there's nothing to multiply. Build the foundation first.

THE BLUEPRINT

Nine regions. Forty-five protocols. One integrated system.

Start with the non-negotiables: nail sleep and exercise. Then add one protocol from your most limiting brain region. Run it for 30 days before adding another. Build the architecture deliberately, one layer at a time.

"Apply them. The performance is waiting."

KEY TAKEAWAY: Don't try to implement 45 protocols. Start with 7-9 hours of sleep and 3 hours of weekly Zone 2 cardio. Once those are locked, identify your single biggest performance bottleneck from the 9 regions and add one protocol. Stack deliberately. Gains compound.

BACK MATTER

Quick Reference Cards

All 9 regions · 45 protocols · Instant access

01 Prefrontal Cortex — Willpower Engine

Executive control · decisions · discipline

1	**Cold Exposure**	2–3 min cold shower daily; norepinephrine + willpower reps
2	**Meditation**	10 min focused attention; gray matter growth
3	**Decision Batching**	Weekly plan + 3 daily priorities; preserve PFC capacity
4	**Sleep Optimization**	7–9h, consistent schedule, 65°F room
5	**Progressive Discipline**	Daily small discomfort challenge; escalate monthly

02 Anterior Cingulate Cortex — Grit Circuit

Effort tolerance · pain threshold · persistence

1	**HIIT**	2x/week; 4–6 × 30-sec max effort; stay with the discomfort
2	**Mindfulness**	10–20 min open monitoring; observe discomfort without obeying
3	**Goal Rituals**	Daily purpose statement before training; pre-competition ritual
4	**Caffeine Optimization**	200–400mg; 30–60 min pre-performance; cycle regularly
5	**Iron Monitoring**	Ferritin test 2x/year; target 50–100+ ng/mL

03 Basal Ganglia — Habit Automation

Automated behavior · dopamine loops · routine execution

1	**Habit Stacking**	"After X, I will Y" anchoring to existing behaviors
2	**Reward Timing**	Celebrate immediately after each behavior rep
3	**Dopamine Hygiene**	No cheap dopamine before training; protect baseline
4	**Schedule Consistency**	Same time, place, pre-training ritual every session
5	**Tyrosine Diet**	Eggs, meat, seeds in pre-training meal; B vitamins as cofactors

04 Hippocampus — Learning Accelerator

Memory formation · neurogenesis · learning retention

1	**Zone 2 Cardio**	3–4x/week; 30–60 min; 60–70% max HR for BDNF
2	**REM Sleep**	7–9h; no alcohol; protect final sleep cycles
3	**Omega-3/DHA**	1–3g EPA+DHA daily; fish or algae-based supplement
4	**Novel Skill Learning**	One genuinely new skill per quarter
5	**Alcohol Awareness**	No alcohol night before key learning sessions

05 Temporal Lobe — Coaching Receiver

Language processing · coaching uptake · skill storage

1	**Active Listening**	Paraphrase coaching; write key points within 5 min post-session
2	**Music & Rhythm**	BPM-matched training; rhythm drills; metronome work
3	**Verbal Rehearsal**	1–2 technical cue words per key skill; use during practice
4	**Audio Priming**	1–2 hrs/week domain-specific podcasts; write key takeaway
5	**Social Interaction**	2–3 high-quality in-person engagements per week

06 Cerebellum — Precision Engine

Motor precision · muscle memory · movement refinement

1	**Deliberate Practice**	20–30 min high-focus with feedback; 70–80% accuracy target
2	**Proprioception**	Daily balance work; BOSU; agility ladder; 5–10 min warm-up
3	**Visualization**	5–10 min first-person mental rehearsal; before practice or sleep
4	**Cross-Training**	Weekly different movement modality; annual cross-training block
5	**Rest for Consolidation**	Sleep after technical practice; don't stack without recovery

07 Amygdala — Threat Processor

Stress response · pressure performance · fear regulation

1	**Box Breathing**	4-4-4-4; 4–6 cycles pre-performance; daily for vagal tone
2	**Stress Exposure**	Graduated competition simulation; escalate stakes progressively
3	**HRV Management**	Track daily; recover when 10%+ below 7-day rolling average
4	**Journaling**	10 min pre/post competition; name fears, then write capability
5	**Cold/Heat Contrast**	Cold immersion + sauna; 3–4x/week; stay present in discomfort

08 Brain Stem — Foundation

Autonomic regulation · HRV · breathing baseline

1	**Nasal Breathing**	Nasal-only during Zone 1–2; mouth tape for sleep if needed
2	**Neck Mobility**	Daily 5-min neck circuit + suboccipital release
3	**Sleep Schedule**	Same wake time ±30 min, 7 days/week; light anchoring protocol
4	**Magnesium Glycinate**	300–400mg; 60 min before bed; avoid oxide form
5	**Diaphragmatic Breathing**	5 min extended exhale (in 4, out 6–8); post-training + pre-sleep

09 Thalamus — Attention Architect

Sensory filtering · attention gating · sleep spindles

1	**Sleep Quality**	Cool room; no alcohol; no screens 90 min before bed
2	**Sensory Training**	Reaction ball; visual tracking; eyes-closed movement drills
3	**Caffeine Timing**	Delay first dose 60–90 min post-wake; last dose by 1pm
4	**Mindfulness**	Focused attention in mild distraction; attentional breathing
5	**Environmental Design**	Clear workspace; DND mode; single-tab focus; pre-session quiet

BACK MATTER

Recommended Reading

Go deeper with the science. Some book links are Amazon affiliate links — we earn a small commission at no cost to you.

01

Why We Sleep

Matthew Walker

02

The Brain That Changes Itself

Norman Doidge

Neuroplasticity in depth — why the brain remains trainable at any age.

The definitive guide to sleep science — PFC sensitivity, REM consolidation, every brain region covered.

03

Spark

John Ratey

The revolutionary science of exercise and the brain — BDNF, hippocampus, learning, and mood.

04

Atomic Habits

James Clear

Basal ganglia, cue-routine-reward loops, identity-based habit formation. Practical and research-backed.

05

Breath

James Nestor

Nasal breathing, autonomic nervous system, CO2 tolerance — how we breathe and why it matters.

06

Peak

Anders Ericsson

The real science of expertise — deliberate practice, cerebellar motor learning, how champions are made.

07

The Willpower Instinct

Kelly McGonigal

PFC science, self-regulation, ego depletion, and practical willpower tools from Stanford.

08

Mindset

Carol Dweck

Growth mindset, effort-belief, neuroplasticity — the psychology of developing ability over time.

09

Can't Hurt Me

David Goggins

The 40% rule in practice. ACC training through radical discomfort exposure. Raw, real, research-validated.

10

Huberman Lab Podcast

Andrew Huberman

Every neuroscience protocol in this book has been covered in depth. Free. Essential. Start with episodes on focus, sleep, and exercise.

ABOUT

Psyked

The Interactive Brain Performance Platform

Psyked brings the neuroscience of performance to life through an interactive 3D brain model that lets you explore every region, understand its function, and connect it directly to your training.

FREE FEATURES

- Interactive 3D brain model
- Performance protocol library
- Region-specific training guides
- Weekly performance content

OUR MISSION

Close the gap between neuroscience research and athletic performance practice. Make the best science accessible, visual, and immediately applicable.

Explore psyked.app — Free

What’s Your Brain Performance Profile?

You've read the science. Now find your personal bottleneck — and your biggest opportunity.

Take the Brain Performance Profile Quiz — $17

Explore the Interactive 3D Brain — psyked.netlify.app →

🔗 Loved this book? Share it — every athlete you send it to gets the same edge you now have.

Follow Psyked: @psykedapp on Instagram, Twitter/X, and YouTube

Join the newsletter: psyked.app/newsletter — weekly brain performance insights

Brain Performance Blueprint — © 2026 Psyked — psyked.app

9 Brain Regions. 45 Protocols. Unlimited Performance.